PROJECT (1)

Coffee Bean Sales

Goal	Practice data analysis & visualization with Python
Tools	Python, Pandas, NumPy, Matplotlib, Seaborn
Dataset	Coffee Bean Sales Raw Dataset
Final Output	Cleaned dataset + Full analysis report

Welcome to Your First Project

Congratulations! You have just been hired as a **Junior Data Analyst** at a specialty coffee bean company. The company has been selling premium coffee beans for several years but has never properly analyzed its sales data. That job is now yours.

You will receive raw, messy data and turn it into actionable business insights. Along the way, you will practice everything you have learned about Python, Pandas, NumPy, Matplotlib, and Seaborn.

The Dataset

Download the **Coffee Bean Sales Raw Dataset** from Kaggle. It contains three related files:

Orders File

- Transaction records of every coffee bean purchase
- Contains order IDs, dates, customer references, product references, quantities

Customers File

- Profile details for each customer
- Includes names, email addresses, locations, loyalty card status

Products File

- Product catalog for the coffee beans
- Includes coffee types, roast levels, sizes, unit prices, profit margins

What You Will Deliver

By the end of this project, you must submit the following:

1. **A cleaned CSV file**, your final cleaned dataset
2. **A Jupyter Notebook**, containing all your code and analysis
3. **A written report**, summarizing findings and recommendations

Project Roadmap

This project is divided into six phases. Each phase builds on the previous one.

Phase	Focus
1	Project Setup & Data Understanding
2	Data Quality Assessment
3	Data Cleaning & Preparation
4	Exploratory Data Analysis
5	Data Visualization
6	Final Report & Presentation

PHASE 1	Project Setup & Data Understanding

Goal

Set up your workspace and get to know your data **before** changing anything.

What You Need To Do

1. Create an organized project folder with subfolders for raw data, cleaned data, notebooks, and outputs
2. Load all three data files into separate DataFrames
3. Inspect the structure of each file (rows, columns, types)
4. Write a short description of each file in your own words
5. Identify how the three files are related to each other
6. Document every observation in markdown cells inside your notebook

Questions You Should Be Able To Answer

- How many rows and columns does each file have?
- What does each column represent?
- Which columns are used to connect the three files?
- What are your first impressions of the data's quality?

What To Submit For This Phase

> **Phase 1 Deliverable**
>
> - A notebook section titled 'Data Understanding'
> - Code showing how you loaded and inspected the data
> - Written observations in markdown cells

PHASE 2	Data Quality Assessment

Goal

Before fixing anything, find every problem in the data and document it.

What You Need To Investigate

Missing Values

- Which columns contain missing values?
- How many values are missing in each column?
- What percentage of the column is missing?

Duplicate Records

- Are there any duplicate rows in any of the tables?

Data Types

- Is each column stored in the correct data type?
- Are there numbers stored as text? Dates stored as plain strings?

Inconsistent Formatting

- Are there leading or trailing spaces in text columns?
- Are there mixed capitalizations for the same value?
- Are there multiple ways the same value is written?

Invalid Values & Outliers

- Are there values that do not make logical sense?
- Are there numeric values far outside the normal range?
- Are there columns where numbers are mixed with symbols or units?

What To Submit For This Phase

Phase 2 Deliverable

- A notebook section titled 'Data Quality Report'
- A written list of every problem you found, grouped by table
- A proposed strategy for fixing each problem, with your reasoning

PHASE 3	Data Cleaning & Preparation

Goal

Execute the cleaning plan you built in Phase 2 and produce a cleaned dataset that is ready for analysis.

What You Need To Do

1. Handle missing values, choose an appropriate strategy for each column and justify it
2. Remove duplicate records where applicable
3. Convert columns to their correct data types (dates, numbers, categories)
4. Standardize text fields so values are consistent
5. Handle outliers
6. Create new columns through feature engineering (date components, calculated fields, categorical bins)
7. Merge the three cleaned tables into one DataFrame
8. Save the unified DataFrame as a CSV file

Code Quality Requirements

- Each major cleaning step must be wrapped in a reusable function
- Running the notebook from top to bottom must always produce the same output
- Every decision must be explained in a markdown cell next to the code
- Show before-and-after comparisons for each cleaning step

What To Submit For This Phase

Phase 3 Deliverable

- A notebook section titled 'Data Cleaning'
- A final cleaned CSV file saved to the outputs folder
- Written justifications for every cleaning decision

Important: *The cleaned CSV you produce here will be the input for Project 2. Take the time to do it properly.*

PHASE 4	Exploratory Data Analysis

Goal

Use Pandas to explore the cleaned data and answer real business questions.

Business Questions You Must Answer (at least 10 of them)

Sales Performance

- What is the total revenue across the entire time period?
- How has revenue evolved over the years?
- Which months and quarters generate the highest sales?
- Are there seasonal patterns in the data?

Product Analysis

- Which coffee types generate the most revenue?
- Which coffee types sell in the largest quantities?
- How does the roast type affect sales performance?
- Which product sizes are most popular?
- Is there a relationship between price and sales volume?
- Which products have the highest profit margins, and are they the best-sellers?

Customer Analysis

- How many unique customers are there?
- Which countries and cities contribute the most to sales?
- What percentage of customers have a loyalty card?
- Do loyalty card holders spend more than non-loyalty customers?
- Who are the top 10 customers by total spending?
- What is the average order value per customer segment?
- How many customers made only a single purchase?

Time-Based & Statistical Analysis

- How does sales performance differ across days of the week?
- What are the descriptive statistics (mean, median, std) for key metrics?
- What is the distribution of order values — is it normal or skewed?

What To Submit for This Phase

Phase 4 Deliverable

- A notebook section titled 'Exploratory Data Analysis'
- Every question answered with code, output, and written interpretation
- Organized by topic (sales, products, customers, time)

PHASE 5	Data Visualization

Goal

Turn your findings into clear, professional charts using Matplotlib and Seaborn.

Minimum Required Visualizations (10 charts)

1. A time-series chart showing revenue over the full period
2. A comparison chart across coffee categories
3. A chart showing geographic distribution of sales
4. A chart examining the relationship between two numeric variables
5. A chart showing the distribution of a continuous variable
6. A chart comparing loyalty vs. non-loyalty customers
7. A chart showing the top performers in some category
8. A heatmap showing patterns across two dimensions
9. A multi-chart figure with subplots combining related views
10. One additional chart of your own choosing

Note: *You can come with your own visualizations (don't stick with those only)*

Quality Standards Every Chart Must Meet

- Clear and descriptive title
- Legend when multiple series are shown
- Saved as a separate image file in the outputs folder
- Written interpretation embedded beside the chart in the notebook

What To Submit For This Phase

Phase 5 Deliverable

- A notebook section titled 'Data Visualization'
- At least 10 charts with interpretations
- All charts saved as image files in the outputs folder

PHASE 6	Final Report & Presentation

Goal

Turn your technical work into deliverables that a non-technical manager could understand and act on.

Written Report

Your report must include:

1. **Executive Summary**, the most important findings in a few bullets
2. **Methodology**, a simple explanation of how you cleaned and analyzed the data
3. **Findings**, organized by theme, supported by charts
4. **Recommendations**, 3 to 5 actionable suggestions for the business
5. **Conclusion,** a short wrap-up

What To Submit for This Phase

Phase 6 Deliverable

- Final written report (PDF or Markdown)
- Complete Jupyter Notebook with all phases documented